Data Project: Closer Look at Toronto's Real Estate

In this project, I'm focusing on evaluating the Toronto real estate market, specifically exploring whether there's a potential asset price bubble. My primary approach involves utilizing linear regression analysis to dissect market dynamics and scrutinize concerns surrounding a possible bubble. I'll be digging into historical data and relevant metrics such as price trends, market volatility, and supply-demand dynamics. By delving into these aspects, I aim to uncover any signals or patterns that might suggest the presence of an asset price bubble in the Toronto real estate market. This analysis will provide me with valuable insights into the market's current state, allowing me to make informed decisions and predictions about its future direction.

Rising bond prices signal lower yields and a safe investment, prompting investors to move funds from stocks, including real estate-linked ones, potentially reducing property demand. Conversely, when stock prices rise, investor confidence grows, driving capital toward real estate, possibly inflating prices. So to see this correlation. The dependent value in this case is bonds. While The TSX being independent. The output from the regression analysis reveals important relationships between the variables.

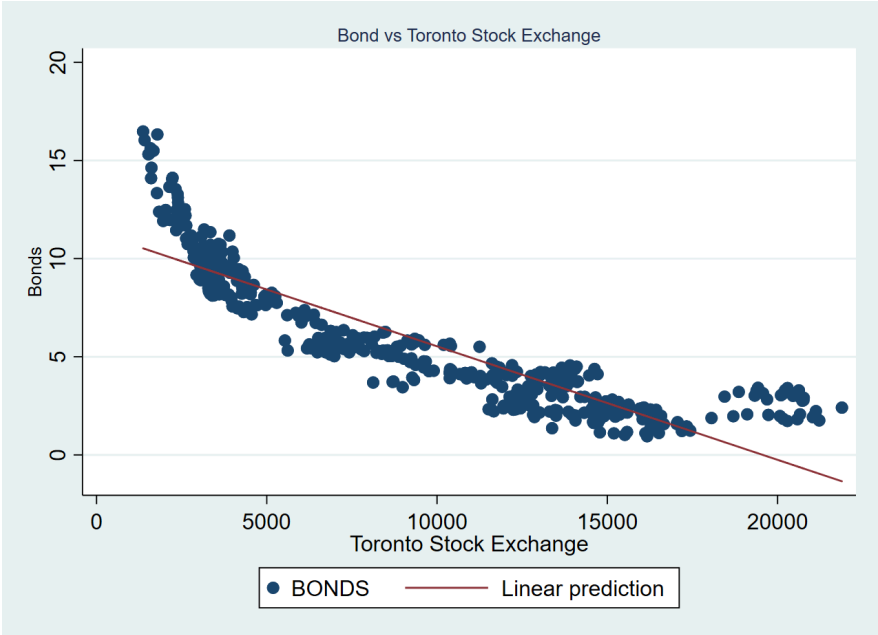
Source	SS	df	MS
Model	4907.47949	1	4907.47949
Residual	1223.53698	498	2.45690157
Total	6131.01648	499	12.2866062

Number of obs	=	500
F(1, 498)	=	1997.43
Prob > F	=	0.0000
R-squared	=	0.8004
Adj R-squared	=	0.8000
Root MSE	=	1.5675

BONDS	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
TSX	-.0005784	.0000129	-44.69	0.000	-.0006038	-.000553
_cons	11.3182	.1373631	82.40	0.000	11.04831	11.58808

With a high F-statistic of 1997.43 and a tiny p-value of around 0.000, the model is highly significant, indicating that at least one of the independent variables, and likely both, have a significant impact on the dependent variable. The substantial R-squared value of 0.8004 suggests that approximately 80.04% of the variation in bond values can be explained by the included independent variables.

Specifically, the coefficient for "TSX" is -0.0005784, indicating a negative relationship with bonds, meaning that as the Toronto Stock Exchange (TSX) value increases, bond values tend to decrease. Both coefficients have extremely low p-values (0.000), confirming their statistical significance. The confidence intervals for these coefficients further support their significance, as they do not include zero.



The negative correlation between bonds and TSX implies a dynamic relationship where investors may shift between these markets based on prevailing economic conditions. This shift in capital allocation can have cascading effects on the real estate sector, influencing its liquidity and overall health.

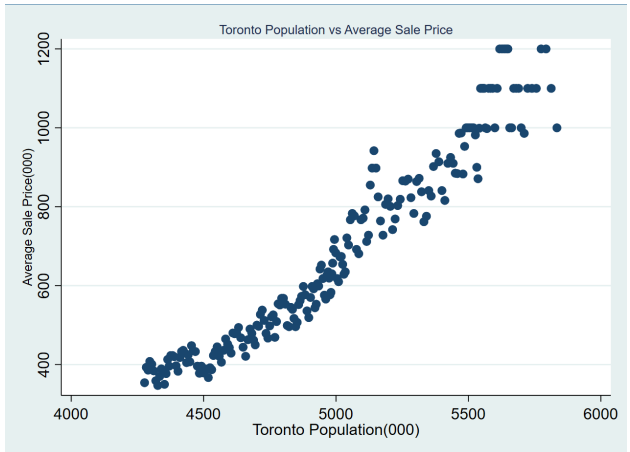
Following the market I will now focus on the population growth and how dependent the variable for the Average Sale Price in Toronto .

Source	SS	df	MS
Model	11938591.7	1	11938591.7
Residual	835229.546	210	3977.28355
Total	12773821.3	211	60539.4374

Number of obs	=	212
F(1, 210)	=	3001.69
Prob > F	=	0.0000
R-squared	=	0.9346
Adj R-squared	=	0.9343
Root MSE	=	63.066

AVGPT	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
POP	.5557924	.0101445	54.79	0.000	.5357944	.5757905
_cons	-2094.177	50.85424	-41.18	0.000	-2194.427	-1993.927

Based on the regression analysis, it's evident that there's a strong relationship between the independent variable (population growth) and the dependent variable (Average Sale Price) in Toronto. With an F-statistic of 3001.69 and a p-value of 0.0000, the overall model is statistically significant.The high R-squared value of 0.9346 suggests that a significant portion of the variability in the Average Sale Price in Toronto can be attributed to changes in population growth. In other words, as the population of Toronto grows, it tends to drive up housing prices, as seen clearly in this upward trend below.



Additionally, the intercept term is -2094.177 with a standard error of 50.85424, and it is also statistically significant. This indicates that even with no population growth, there is a base Average Sale Price in Toronto. Overall, these findings emphasize the importance of population growth in understanding fluctuations in the Average Sale Price in Toronto.

On a different point of view we could examine population dynamics in relation to Toronto's price index for condos and two-storey houses through multiple regression analysis that yields valuable insights into their interrelationship:

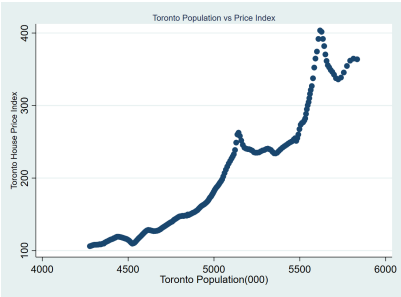
Source	SS	df	MS
Model	35229467.2	3	11743155.7
Residual	3418607.58	208	16435.6134
Total	38648074.8	211	183166.231

Number of obs	=	212
F(3, 208)	=	714.49
Prob > F	=	0.0000
R-squared	=	0.9115
Adj R-squared	=	0.9103
Root MSE	=	128.2

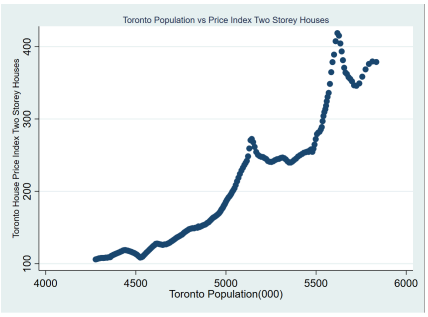
POP	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
THPI	8.473595	11.66305	0.73	0.468	-14.51935	31.46654
THPIH	-4.905725	10.33473	-0.47	0.636	-25.27997	15.46852
THPIA	1.671421	.9604669	1.74	0.083	-.2220769	3.564919
_cons	3972.044	68.53208	57.96	0.000	3836.937	4107.15

Firstly, the statistical significance of the model is evident, with an F-statistic of 714.49 and a p-value of 0.0000, indicating that the independent variables collectively have a significant impact on the dependent variable. Furthermore, the high R-squared value of 0.9115 suggests that approximately 91.15% of the variability in Toronto's price index can be explained by the independent variables, namely THPI, THPIH, and THPIA.

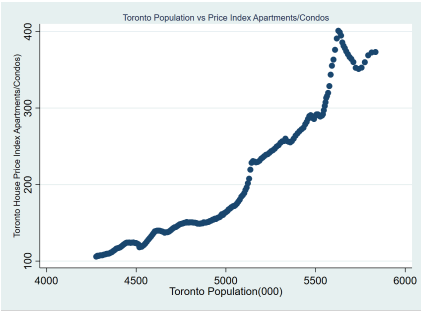
THPI



THPIH



THPIA



	THPI	THPIH	THPIA	POP
THPI	1.0000			
THPIH	0.9999	1.0000		
THPIA	0.9821	0.9794	1.0000	
POP	0.9517	0.9505	0.9489	1.0000

However, upon closer examination of the coefficients and the correlation matrix provides insight into the relationships between variables. For instance, it shows strong correlations between THPI and THPIH (0.9999), THPI and THPIA (0.9821), and THPI and POP (0.9517). Only the coefficient for THPIA appears to be marginally statistically significant at a 10% significance level, with a p-value of 0.083. This implies that Toronto's price index for all home types (THPIA) may have some influence on the overall price index, while the other variables may not be significant predictors.

In exploring the relationship between population and Toronto's price index for condos and two-storey houses,we see population growth influences the demand for these types of homes and, in turn, affects their prices.

We will next integrate Toronto's Labor Force Composition (LFC), Part-time Employment (PTM), Unemployment (UNEMP), and Employment Insurance (EI) metrics with Real Estate indicators for Toronto Average Sale Price, Houses & Condos(AVGPT). This integration is crucial as employment and economic factors significantly influence housing demand.

Source	SS	df	MS
Model	11222418.5	4	2805604.63
Residual	1449727.14	206	7037.5104
Total	12672145.7	210	60343.5508

Number of obs	=	211
F(4, 206)	=	398.66
Prob > F	=	0.0000
R-squared	=	0.8856
Adj R-squared	=	0.8834
Root MSE	=	83.89

AVGPT	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
LFC	.8961625	.0343943	26.06	0.000	.8283525	.9639726
PTM	-.1125156	.1641949	-0.69	0.494	-.4362336	.2112023
UNEMP	-.1182468	.1199268	-0.99	0.325	-.3546881	.1181946
EI	.0000474	.000141	0.34	0.737	-.0002305	.0003253
_cons	-2252.744	75.78024	-29.73	0.000	-2402.149	-2103.34

The provided regression analysis output presents essential statistical details regarding the relationship between Toronto's labor force composition (LEC), part-time employment (PTM), unemployment (UNEMP), employment insurance (EI), and the average sale price of houses and condos (AVGPT). With 211 observations, the model demonstrates a robust explanatory power, as indicated by the high F-statistic of 398.66 and an R-squared value of 0.8856, suggesting that approximately 88.56% of the variability in average sale prices can be explained by the included variables.

Upon examining the coefficients, it's evident that the labor force composition (LEC) variable carries significant explanatory weight, with a coefficient of 0.8961625. This coefficient's statistical significance is reinforced by its high t-statistic of 26.06 and a p-value below 0.000, indicating its strong influence on average sale prices. However, the coefficients for part-time employment (PTM), unemployment (UNEMP), and employment insurance (EI) exhibit less significance, as indicated by their respective t-statistics and p-values. Specifically, PTM, UNEMP, and EI coefficients stand at -0.1125156, -0.1182468, and 0.0000474, respectively, with their associated t-statistics failing to exceed conventional thresholds for statistical significance, and p-values above 0.05.

The integration of Toronto's labor force composition, part-time employment, unemployment, and employment insurance metrics with real estate indicators is pivotal, considering the significant influence of economic factors on housing demand. While the labor force composition emerges as a robust determinant of housing prices, the less pronounced impact of part-time employment, unemployment, and employment insurance warrants further investigation to understand their nuanced roles in shaping Toronto's real estate market dynamics.

So we could further look at the correlation between part-time employment, unemployment, and employment insurance to get a deeper confirmation they are not related:

	PTM	UNEMP	EI
PTM	1.0000		
UNEMP	-0.1275	1.0000	
EI	-0.0604	0.5244	1.0000

The correlation matrix reveals insights into the relationships among part-time employment (PTM), unemployment (UNEMP), and employment insurance (EI) variables. While PTM and UNEMP exhibit a weak negative correlation (-0.1275), suggesting a slight inverse relationship, PTM and EI show a very weak negative correlation (-0.0604), indicating a minimal association. Conversely, UNEMP and EI display a moderate positive correlation (0.5244), implying a more pronounced connection between unemployment and employment insurance. Overall, these findings suggest that while there are some interdependencies among the variables, they are not strongly correlated, highlighting the need for further investigation into their individual impacts on Toronto's real estate dynamics.

Given that, let's now examine the most crucial part. The Consumer Price Index for Toronto (CPIT) alongside Real Estate indicators such as Retail trade sales (RETAIL), New House Price Index (CNEWHL), Price of Rented accommodation (RENT), and Price of owned accommodation (OWN) in Toronto and Canada are potential variables that may correlate with the Consumer Price Index (CPIT) due to their influence on inflation and consumer spending patterns. In this case multiple regression analysis can be a valuable tool for analyzing the relationship between variables:

Source	SS	df	MS			
				Number of obs	=	235
				F(4, 230)	=	8990.34
Model	50287.6634	4	12571.9158	Prob > F	=	0.0000
Residual	321.627415	230	1.39838007	R-squared	=	0.9936
				Adj R-squared	=	0.9935
Total	50609.2908	234	216.279021	Root MSE	=	1.1825

CPIT	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
RETAIL	6.27e-07	2.21e-07	2.83	0.005	1.91e-07	1.06e-06
CNEWHL	.0996249	.0229173	4.35	0.000	.0544703	.1447795
RENT	.3228936	.044259	7.30	0.000	.2356886	.4100985
OWN	.4200989	.0342393	12.27	0.000	.352636	.4875618
_cons	17.57293	2.239767	7.85	0.000	13.15985	21.98602

The provided regression analysis demonstrates a strong relationship between variables RETAIL (Retail trade sales), CNEWHL (New House Price Index), RENT (Price of Rented accommodation), OWN (Price of owned accommodation), and the Consumer Price Index (CPIT), indicating their significant influence on inflation and consumer spending patterns. Nearly all variables having p-value of 0.0000(or close to 0) and the high R-squared value of 0.9936 indicates that the model explains 99.36% of the variability in CPIT, suggesting a robust explanatory power. Each coefficient represents the change in CPIT associated with a one-unit increase in the respective predictor variable.

Retail trade sales (RETAIL) positively impact CPIT, indicating that higher retail spending contributes to inflation. Similarly, increases in both the New House Price Index (CNEWHL) and prices of rented (RENT) and owned (OWN) accommodations are associated with higher CPIT, reflecting the significant weight of housing costs in the CPI basket and their contribution to overall inflation.

The strong correlation between these variables and CPIT raises concerns about the potential formation of a price bubble in the Toronto housing market. The significant influence of housing-related variables on inflation suggests that rising housing prices contribute substantially to overall price levels. If left unchecked, this trend may lead to unsustainable increases in housing prices, eventually resulting in a price bubble.

In conclusion, my analysis reveals strong correlations between various economic factors and Toronto's real estate market, suggesting potential risks of a price bubble. Factors such as population growth, labor force composition, and housing costs significantly influence housing prices and inflation. The close relationship between these variables underscores the need for vigilant monitoring and proactive measures to ensure market stability and affordability. Policymakers should prioritize sustainable housing policies to mitigate the risks associated with a potential price bubble and foster a resilient housing market in Toronto.